Kartik Agarwal

FULL-STACK WEB DEVELOPER

New Delhi, India | +91 76684 40798 | agarwalkartik704@gmail.com | linkedin.com/in/kartik-agarwal | github.com/scripter-kartik | kartikagarwal.me

Full-stack developer and third-year CSE undergraduate who ships production-style web applications with Next.js, React, and Node.js. Comfortable across the stack — real-time features over WebSockets, REST API design, authentication, and MongoDB data modeling.

EDUCATION

Vivekananda Institute of Professional Studies – TC (GGSIPU) — Aug 2024 – May 2028

B.Tech, Computer Science & Engineering · New Delhi

Relevant coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Theory of Computation, Probability & Statistics

TECHNICAL SKILLS

Languages	C++, JavaScript (ES6+)
Frontend	React.js, Next.js, Tailwind CSS, HTML5, CSS3
Backend	Node.js, Express.js, REST APIs, Socket.IO
Databases	MongoDB, Mongoose (ODM)
Auth & APIs	NextAuth, Clerk, JWT, Gemini API, Google Books API
Tools	Git, GitHub, Linux (Fedora), VS Code, Postman, Vercel

PROJECTS

BlogForge — Full-Stack Blogging Platform

GitHub · Live

Next.js · MongoDB · NextAuth · Tailwind CSS

- Built a full-stack blogging platform with JWT-based authentication and role-based profile management, backed by REST APIs supporting complete CRUD on posts and users.
- Engineered rich post creation and editing with nested comment threads and replies, modeling the relational content graph in MongoDB using Mongoose schemas.
- Added a trending-posts ranking algorithm and keyword search to surface relevant content and improve discovery.

TuneTogether — Real-Time Collaborative Listening App

GitHub · Live

Next.js · Socket.IO · Clerk

- Built a real-time platform that synchronizes music playback across multiple users in shared rooms, implemented end-to-end on Next.js API routes with no separate backend server.
- Engineered WebSocket synchronization with Socket.IO, holding playback drift to sub-second latency across concurrent sessions.
- Shipped room-based chat and track sharing for live group interaction, with Clerk managing authentication and sessions across devices.

InkShelf — AI Book Discovery & Summaries

GitHub · Live

Next.js · Gemini API · Google Books API · Tailwind CSS

- Built a responsive book-discovery platform that generates AI-powered summaries with the Gemini API over catalog data from the Google Books API.
- Reduced redundant network requests through debounced search and optimistic-UI loading skeletons, improving perceived speed and lowering server load.
- Migrated the catalog source from OpenLibrary to the Google Books API to restore availability after the original endpoint was blocked.

EXPERIENCE

Aurum Protocol — Strategy Intern

Remote · May – Jun 2025

- Contributed product and UX feedback for an early-stage startup, shaping feature direction alongside the founding team.
- Ran research and analysis supporting product growth and go-to-market planning in a fast-paced, cross-functional environment.